



EXCEL TRANSFER TOOL

TECHNICAL DOCUMENTATION & USER GUIDE



VERSION

2.0

April 3, 2026

Qt 6

C++17

OpenXML



Finance Automation Team

Internal Use Only · Confidential Document



Excel Transfer Tool is a high-performance Qt/C++ desktop application that automates monthly cost-control reporting workflows. It loads Excel workbooks (XLSM/XLSX) for multiple periods, applies configurable transfer mappings, consolidates data across months, and produces clean output files—all without manual copy-paste work.

Key Capabilities

- Load multiple cost-control files with parallel processing (2-3x faster)
- Apply row-level transfer mappings per month/sheet/column
- Rolling Transfer (RT) mode for sequential month-to-month workflows
- Support for SAP, SAP YTD, PAX, Staff, Budget, and REFI data sources
- Quarter quick-select buttons (Q1-Q4) for rapid period selection
- Batch cell updates with reduced lock contention
- Automatic formula recalculation and external link cleanup
- Export multiple selected months into consolidated output

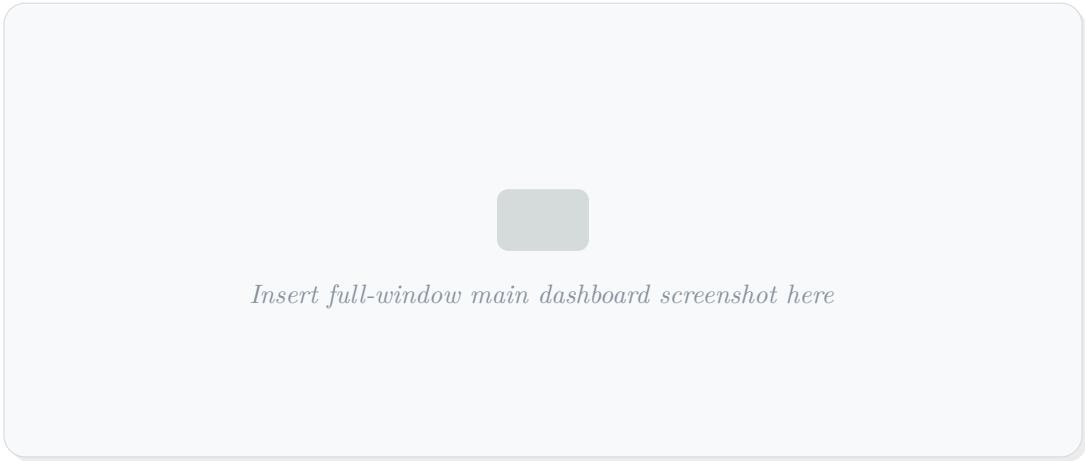


Figure 1: Application Main Dashboard

System Requirements

Component	Requirement
Operating System	Windows 10 or Windows 11 (64-bit)
Microsoft Excel	Excel 2016, 2019, 2021, or Microsoft 365
Runtime	Qt 6.x (bundled with executable)
Compiler	Built with MSVC 2019+ / MinGW (C++17)
Disk Space	Minimum 200 MB free
RAM	8 GB minimum, 16 GB recommended for 12+ months

Setup Steps

1. Extract the application folder to your preferred location (e.g. Desktop).
2. Launch `ExcelTransferTool.exe`.
3. On first launch, verify the default folder paths point to your network share (L: drive).
4. Ensure cost-control files follow the expected naming convention (see Appendix).
5. Verify JSON mapping files exist in the `JSON/` subfolder.

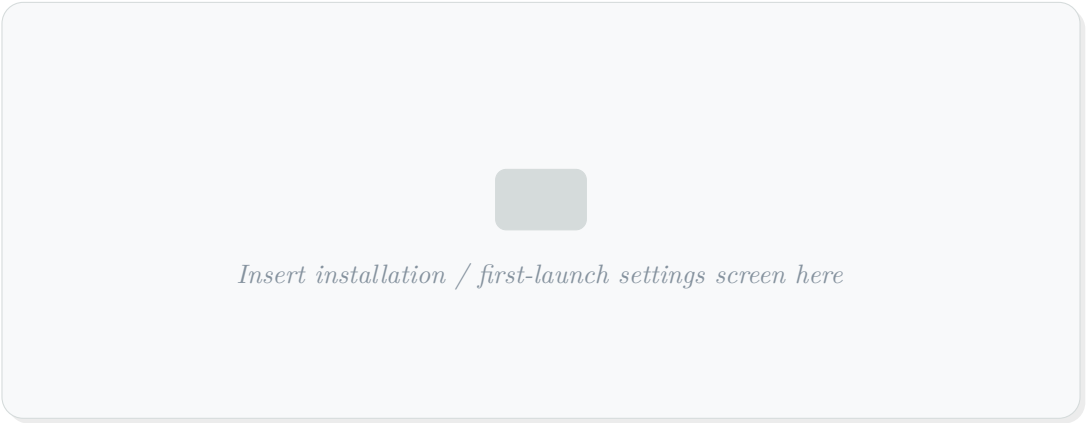


Figure 2: First-Launch Settings Screen



1. **Select years** by checking year checkboxes (2024, 2025, 2026, etc.).
 2. Click **Generate Period Rows** to create month selection cards for each year.
 3. **Select months** by checking individual months or using Q1-Q4 quick-select buttons.
 4. Click **LOAD Selected Periods** to load workbooks (uses parallel loading for speed).
 5. Review the mapping cards that appear in the Active Mappings section.
 6. Choose an action:
 - **Execute All** — runs all checked mapping cards in sequence.
 - **Execute RT** — runs Rolling Transfer for sequential month processing.
 - **Export Selected Months** — consolidates multiple months into one file.
 7. Monitor progress in the status bar and progress indicator.
 8. When complete, verify output files in the destination folder.

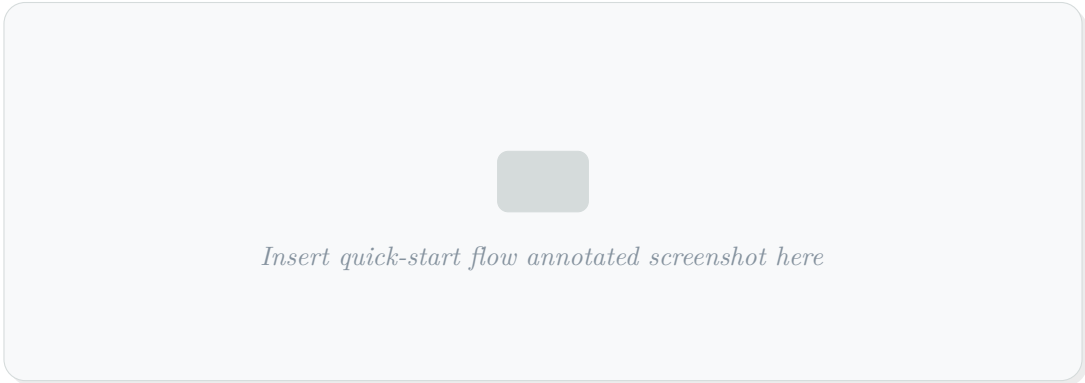


Figure 3: Quick Start Workflow



Selecting Periods

Use the year checkboxes and month selection cards to choose periods. The application supports:

- **Individual month selection** — check specific months
- **Quarter quick-select** — Q1 (Jan-Mar), Q2 (Apr-Jun), Q3 (Jul-Sep), Q4 (Oct-Dec)

- **Quarter removal** — uncheck entire quarters at once
- **Multi-year selection** — select months across multiple years simultaneously

Tip

Use the **Q1–Q4 quick-select buttons** at the top of the period selector to apply or remove entire quarters across all generated period rows.

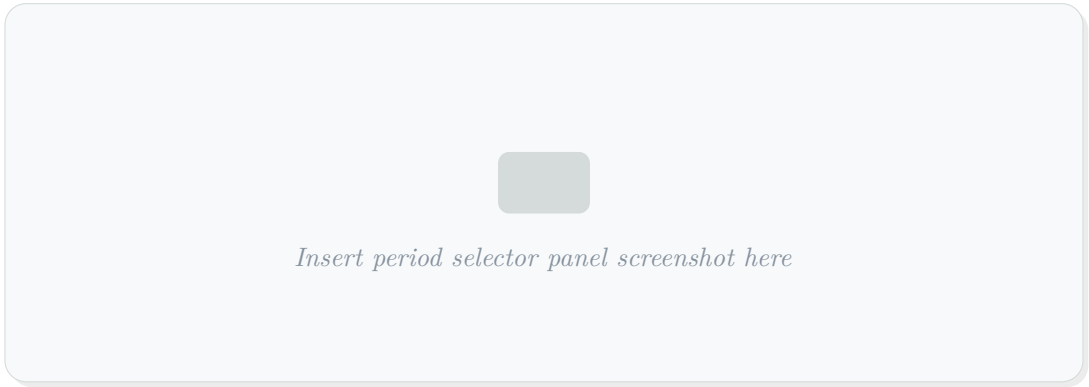


Figure 4: Year and Month Period Selection

Loading Workbooks

Click **LOAD Selected Periods** to open matching cost-control files. The application uses:

- **Parallel loading** with thread pool (3 concurrent files)
- **Lazy sheet parsing** — only loads sheets when accessed
- **ZIP entry caching** — prevents re-reading network files
- **Progress updates** — real-time feedback during load

Important

Ensure all target files are **closed in Excel** before loading. Open files may cause read errors or file locking issues.

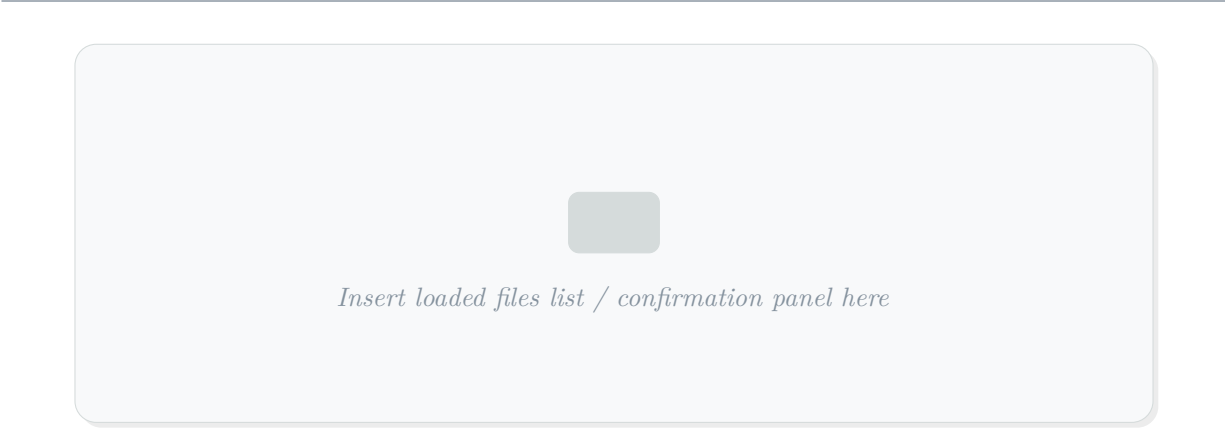


Figure 5: Loaded Files Confirmation Panel



What is a Mapping Card?

Each mapping card represents a transfer rule between:

- A **source** workbook, sheet, and column (where data is read from)
- A **destination** sheet and column (where data is written to)
- A set of **source rows** and their corresponding **destination rows**
- Optional **row mapping** (rowMap) for summing multiple source rows into one destination

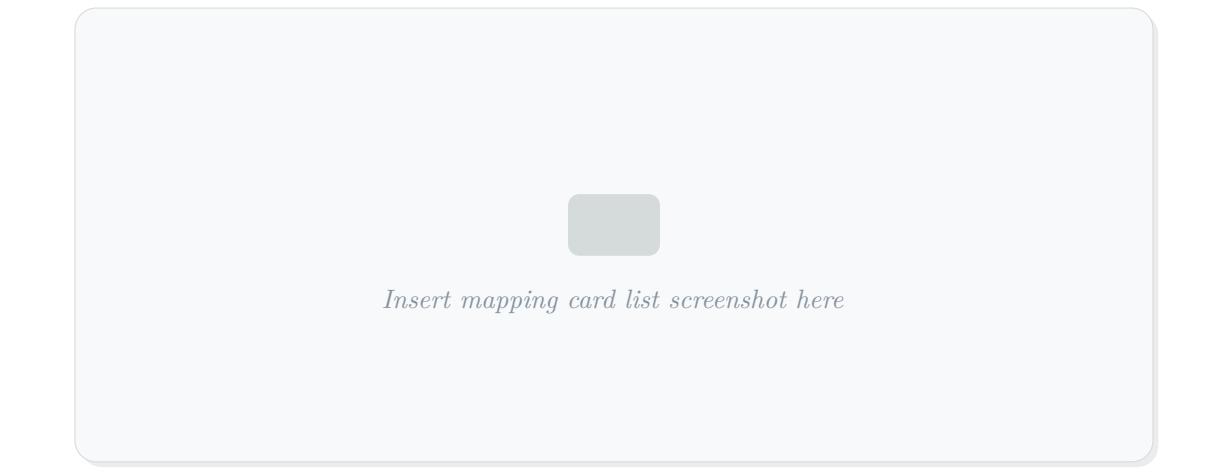


Figure 6: Mapping Cards Overview

Data Source Types

The application supports multiple data source types:

Source Type	Description
SAP	Monthly SAP export files (MM_YYYY.xlsx)
SAP YTD	Year-to-date SAP cumulative data
PAX	Passenger traffic data (monthly columns)
Staff	FTE staffing data (annual sheet per year)
Cost Control	Same-file transfers within cost control workbook
Budget	Annual budget reference columns
REFI	Re-forecast columns for current or prior year

Editing Row Mappings

Click **Edit Rows** on any card to:

- Adjust individual row numbers
- Add new source/destination row pairs
- Remove unused rows
- Configure rowMap for sum operations

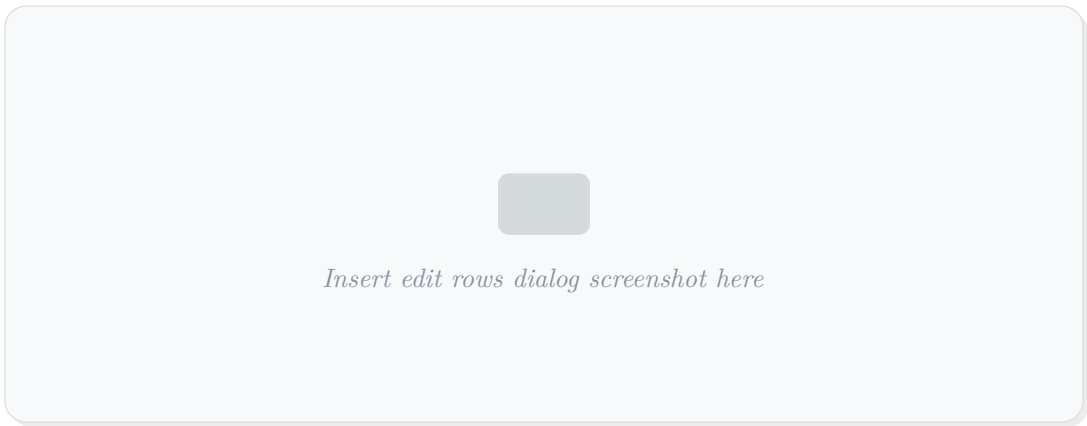


Figure 7: Edit Rows Dialog

Select All / Deselect All

Use the **Select All / Deselect All** toggle button to quickly check or uncheck all mapping cards. Only checked cards will be executed when you click **Execute All**.

What is Rolling Transfer?

Rolling Transfer is a specialized workflow that processes months sequentially, where each month's output becomes the input for the next month. This is used when:

- You need to "roll forward" data from one month to the next
- Each month builds upon the previous month's calculations
- You want to create a chain of monthly files in sequence

How RT Works

1. Select multiple months in chronological order
2. Click **Load RT** to validate the chain:
 - Verifies previous month file exists for the first month
 - Warns if destination files already exist
 - Loads all required mappings
3. Click **Execute RT** to run the chain:
 - Copies previous month's file to new month's test/ folder
 - Loads the copied file as destination
 - Applies all mappings for that month
 - Saves and moves to next month

RT Validation

RT validates that the previous month's file exists before starting. If January 2026 is your first month, the December 2025 file must exist in the source folder.

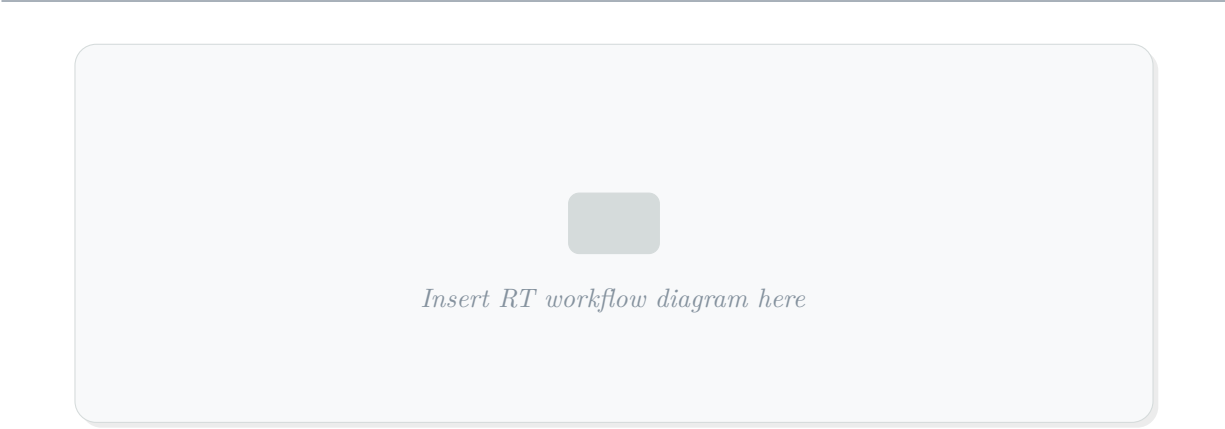


Figure 8: Rolling Transfer Workflow



How It Works

Export Selected Months consolidates data from multiple months into a single output workbook (the **latest selected month** file). This is used when you need to deliver a period report covering multiple months.

The process:

1. Reads each selected month’s source file
2. Transfers all mapping rows into the latest month’s workbook
3. Saves the combined file to the configured destination folder
4. Cleans all external links and triggers recalculation

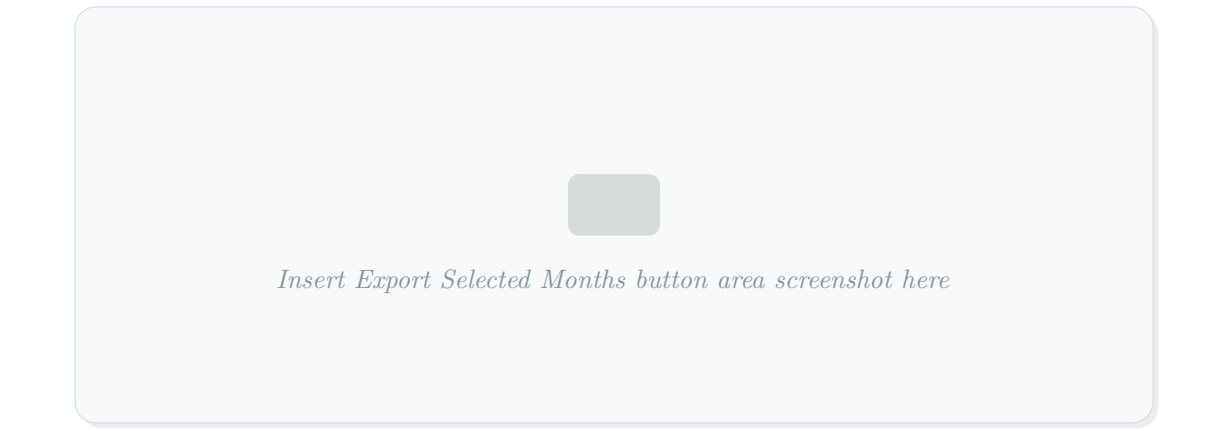


Figure 9: Export Selected Months Button

Running the Export

1. Select all months you wish to consolidate
2. Click **LOAD Selected Periods** to load the workbooks
3. Click **Export Selected Months**
4. Monitor progress in the status bar
5. Verify the output file in the destination folder

Automatic Mapping Inclusion

The export automatically includes all mapping types (SAP, SAP YTD, PAX, Staff, Budget, REFI) for all loaded months, as long as the corresponding source workbooks are available.

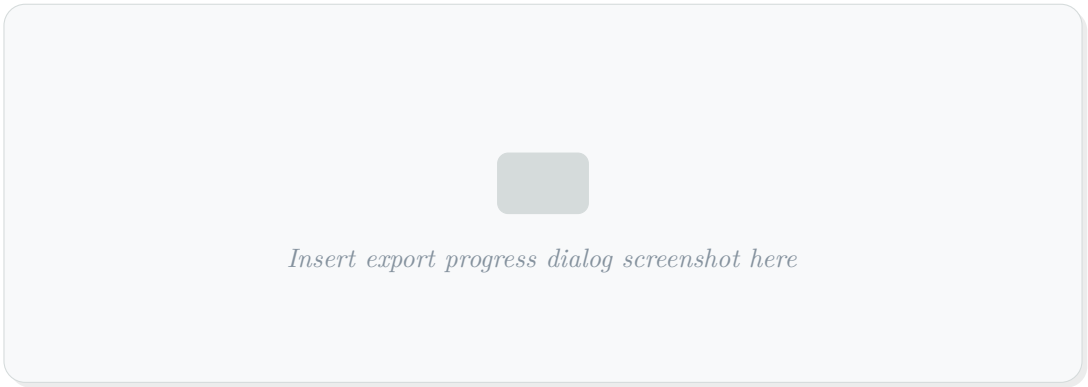


Figure 10: Export Progress and Completion

Running All Transfers

Execute All processes every checked mapping card in order. It reads each source value and writes it to the destination workbook using batch cell updates for optimal performance.

1. Ensure files are loaded
2. Check the mapping cards you want to execute (or use Select All)
3. Click **Execute All**
4. Monitor progress in the status bar and progress indicator
5. On completion, a summary shows total cells transferred

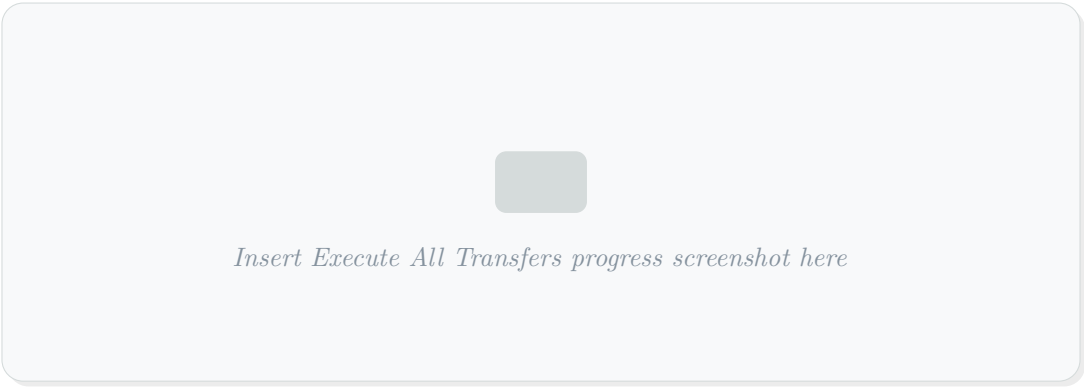


Figure 11: All Transfers Running

Pause and Stop

You can interrupt a running transfer using:

- **Pause** — temporarily halts execution, can be resumed
- **Stop** — terminates execution immediately

Warning

Stopping a transfer mid-run may leave some destination cells partially updated. Always review the output after an interrupted transfer.

Performance Optimizations

The application uses several optimizations for fast execution:

- **Batch cell updates** — groups multiple cell writes into single operations
- **Reduced lock contention** — minimizes thread synchronization overhead
- **In-memory processing** — all operations happen in RAM before saving
- **Lazy sheet loading** — only parses sheets when accessed

After each transfer or export, the application automatically:

- 1. **Writes patch file** to local temp folder using miniz compression
- 2. **Verifies patch file** size and integrity before touching original
- 3. **Creates backup** by renaming original to .bak
- 4. **Copies patch** to final destination (network share safe)
- 5. **Removes backup** only after successful copy
- 6. **Preserves timestamps** for all ZIP entries
- 7. **Stores metadata** uncompressed ([Content_Types].xml, .rels files)

Finalization Complete

When finalization completes, a green status message confirms the output path and the number of cells transferred. The .bak file is automatically removed on success.

File Safety

If the copy operation fails, the original .bak file is automatically restored, ensuring no data loss. The patch file is left in the temp folder for inspection.

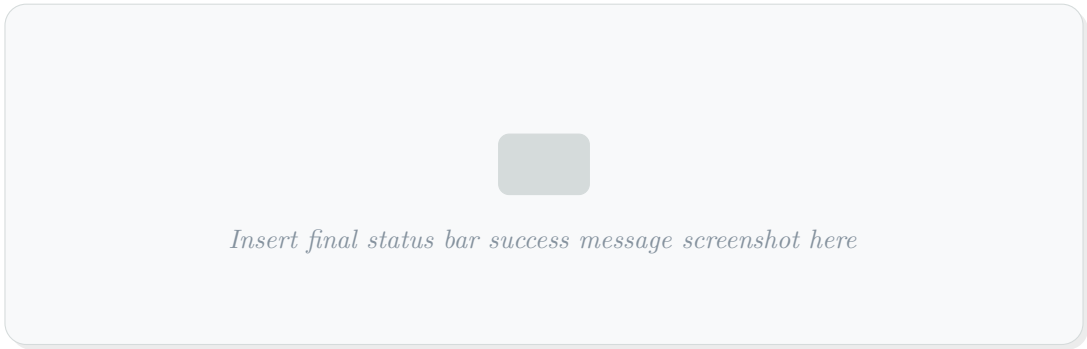


Figure 12: Finalization Success Message

Files Not Saving

Symptom

Execute All completes but output files are unchanged or missing.

Checklist:

- Check debug log for "miniz finalize failed" or "patch file missing"
- Verify destination file is **closed in Excel**
- Ensure temp folder (C:\Users\{user}\AppData\Local\Temp) is writable
- Check disk space on both temp drive and network share
- Look for .bak files — if present, copy failed and original was restored

Empty Cells After Export

Symptom

Cells that should contain SAP, PAX, or Staff values appear blank after export.

Checklist:

- Confirm destination file is **closed in Excel** before exporting
- Verify each month's source files (SAP, PAX, Staff) are available and loaded
- Check the debug console for "file missing" or "sheet not found" messages
- Ensure JSON mapping files have correct sheet names (case-sensitive)

Slow Loading Performance

- Verify network connection to L: drive is stable
- Check if other users are accessing the same files
- Consider reducing thread pool size in loadworker.cpp (line 50) from 3 to 2
- Close unnecessary applications to free RAM

File Not Found

- Verify source folder path points to correct network location
- Confirm file follows naming convention: `Cost Control ZAG MM_YYYY_working.xlsx`
- Check if file is in test/ subfolder vs root folder
- Ensure file has not been moved, renamed, or locked by another user

Mapping Card Missing

- Open the relevant JSON file (mappings.json, mappings_old.json, pax.json, staff.json, etc.)
- Confirm the target month/year entry exists
- Verify sheet names match exactly (case-sensitive)
- Check that `source_file_type` is correctly set (sap, sap_ytd, pax, staff, cost_control)

RT Validation Fails

- Ensure previous month's file exists (e.g., Dec 2025 for Jan 2026 RT)
- Check that months are selected in chronological order
- Verify destination test/ folders exist or can be created
- Review RT validation messages in the status bar

Q: Can I export more than two months into a single file?

A: Yes. Select all desired months, load them, then click **Export Selected Months**. All months are consolidated into the latest month's file.

Q: What's the difference between Execute All and Execute RT?

A: **Execute All** processes all checked mappings independently. **Execute RT** processes months sequentially, where each month's output becomes the next month's input.

Q: Do I need to keep Excel open while the app runs?

A: No. The app handles file I/O directly using OpenXML (ZIP) parsing. Excel is never launched during operation.

Q: Can I customize which rows are transferred?

A: Yes. Use the **Edit Rows** button on each mapping card to adjust source and destination row numbers, or edit the JSON files directly.

Q: How do I add a new mapping type?

A: Create or edit the appropriate JSON file in the JSON/ folder. The app automatically loads all JSON files on startup. Restart the app after editing JSON files.

Q: Why is parallel loading limited to 3 files?

A: Large XLSM files (10-50MB) cause memory pressure when loaded in parallel. The thread pool limit of 3 balances speed with stability. This can be adjusted in loadworker.cpp if you have more RAM.

Q: What happens to external links in the output file?

A: External links are automatically broken after each export to ensure the output file is self-contained and opens cleanly in Excel.

File Naming Convention

Cost-control files must follow this naming pattern:

```
Cost Control ZAG {MM}_{YYYY}_working.xlsm
```

Examples:

Month	Filename
January 2026	Cost Control ZAG 01_2026_working.xlsm
June 2026	Cost Control ZAG 06_2026_working.xlsm
August 2026	Cost Control ZAG 08_2026_working.xlsm
December 2026	Cost Control ZAG 12_2026_working.xlsm

SAP Files:

```
{MM}_{YYYY}.xlsx
```

PAX Files:

```
TRAFFIC mott {YYYY}.xlsx
```

Staff Files:

```
{YYYY} FTE.xlsx
```

Mapping JSON Structure

Standard Mapping (mappings.json)

```
{
  "source_sheet": "{year}",
  "source_column": "C",
  "dest_sheet": "MZLZ Consolidated",
  "rowMap": {
    "10": 5,
    "11": [6, 7]
  },
  "month_mappings": [
    {"month": "January", "source_column": "C", "dest_column": "G"},
    {"month": "February", "source_column": "C", "dest_column": "W"}
  ]
}
```

PAX Mapping (pax.json)

```
{
  "source": {
    "sheet": "TRAFFIC mott 2025",
    "source_rows": [19, 34, 81]
  },
  "dest": {
    "sheet": "MZLZ Consolidated",
    "dest_rows": [5, 7, 8]
  },
  "month_mappings": [
    {"month": "January", "source_column": "G", "dest_column": "G"}
  ]
}
```

Staff Mapping (staff.json)

```
{
  "source": {
    "sheet": "{year} FTE",
    "row": 33
  },
  "dest": {
    "sheet": "MZLZ Consolidated",
    "row": 16
  },
  "month_mappings": [
    {"month": "January", "source_column": "D", "dest_column": "G"}
  ]
}
```

Keyboard Shortcuts

Shortcut	Action
Ctrl + L	Load Selected Periods
Ctrl + E	Execute All Transfers
Ctrl + R	Execute Rolling Transfer
Ctrl + X	Export Selected Months
Escape	Stop running transfer
F5	Refresh mapping cards

Technical Architecture

Component	Technology
UI Framework	Qt 6.x (Widgets)
Language	C++17
Compiler	MSVC 2019+ / MinGW
Excel Parsing	OpenXML (ZIP + XML)
Compression	miniz (embedded)
Threading	QThread, QThreadPool, QtConcurrent
JSON Parsing	Qt JSON (QJsonDocument)
File I/O	Qt File API + QZipReader/Writer

Performance Characteristics

Operation	Performance
Load 12 months (sequential)	96 seconds
Load 12 months (parallel)	32 seconds (3x faster)
Transfer 1000 cells (batch)	0.5 seconds
Save 50MB XLSM file	3-5 seconds
RT chain (6 months)	2-3 minutes

Support

For technical issues, feature requests, or mapping configuration help, contact the internal Finance Automation support team.

Excel Transfer Tool

•

User Manual v2.0

Finance Automation Team • Internal Use Only • April 3, 2026